SYSTEM ANALYSIS



CAR DEALER

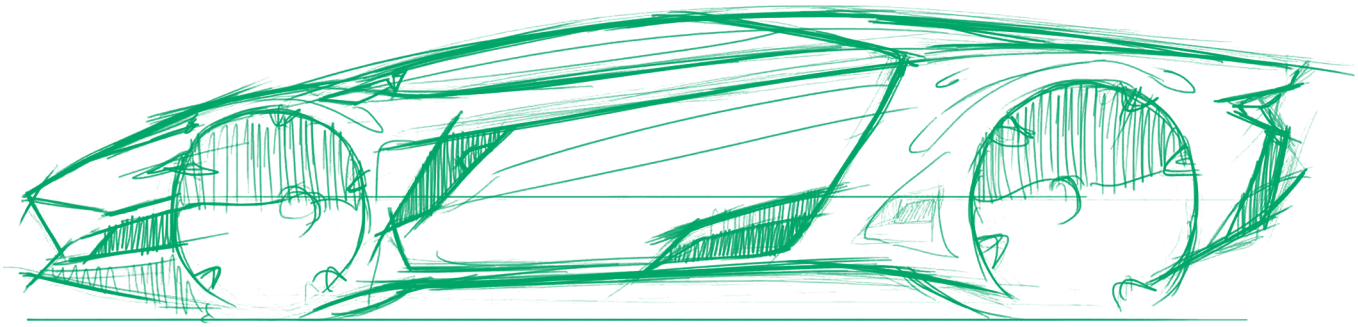
1#

APRIL 2026

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Our Team

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- Chan Punleur
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HOBBY SEARCH

別売:『ランボルギーニアヴェンタドール共通ディテールアップパーツについて』
Optional: Lamborghini Aventador mutual detail up parts.

★このキットには、別売の「ランボルギーニ アヴェンタドール共通ディテールアップパーツ』(001400：1,800円/税別)を組み込むことが可能です。
よりいっそうのディティールアップをお望みの方にお勧めします。
You can attach the "Lamborghini Aventador mutual detail up parts" as option (1,800 yen + tax).
We recommend that to the people who want finer details.

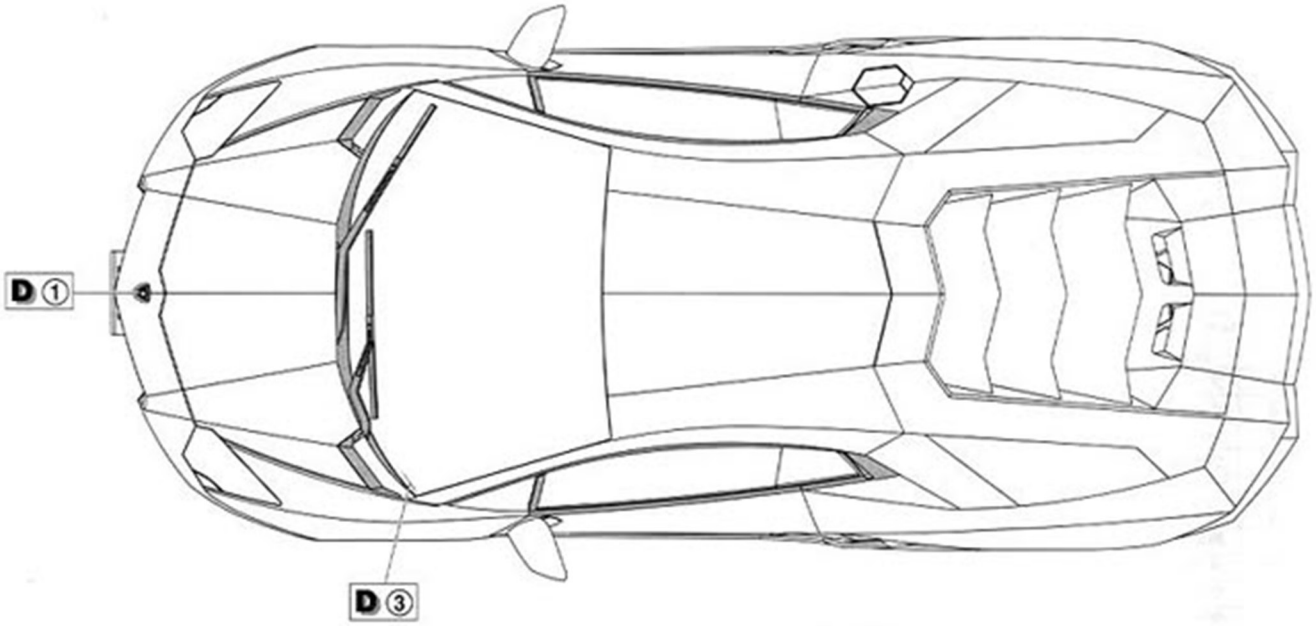
〈 ボディの塗装／Paint instruction for body 〉

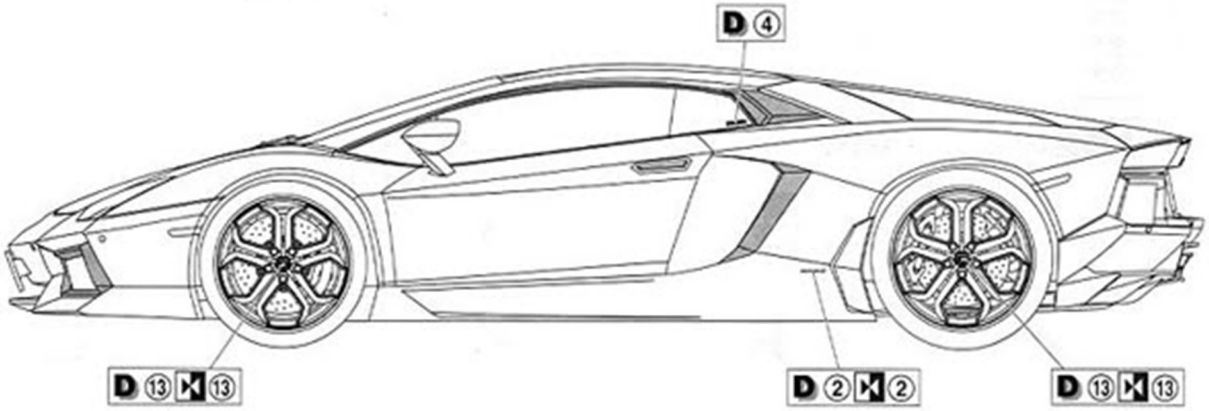
Pはペイントマークです
is Paint Mark.
Zeichen Anstrich.
Marque De Peindre.
是涂颜色的标记

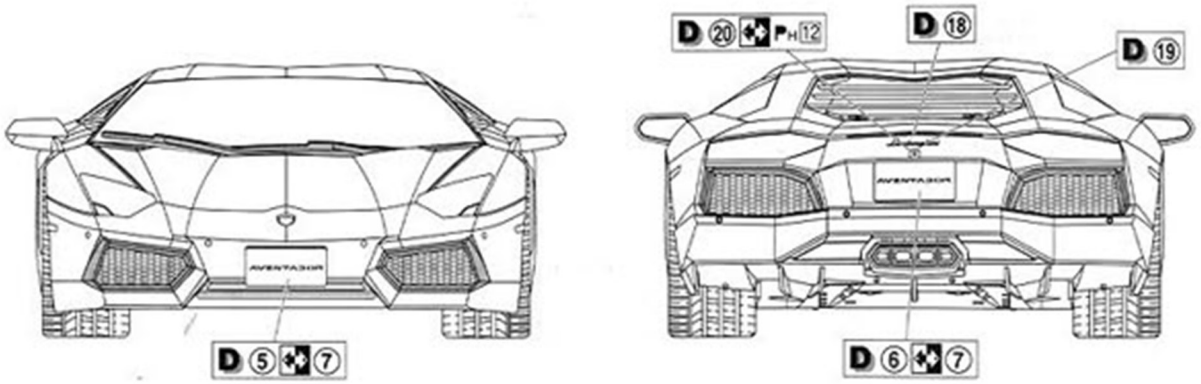
ボディ色
Body color
Karrosseriefarbe
Couleur carrosserie
車体色
PH95(6)+H8(1)→H30

PH2+H48 (少量) (a little)
半光沢のブラックになるように調整します。
Mix colors to semi-gloss black

※デカール⑧～⑪はお好みでご使用下さい。
Decal No. ⑧～⑪ are optional.







Car Dealer System Requirements

1. 🎯 Purpose

The **Online Car Dealer System** allows customers to:

- View available cars
- Check rental/sale prices
- Make reservations online
- Complete payments online

The system aims to provide a **fast, secure, and user-friendly platform** for car browsing and booking.

2. 📤 System Outputs

The system should provide:

- ✅ List of **available cars**
- ✅ **Car details** (model, price, type, availability)
- ✅ **Total price calculation** based on customer selection
- ✅ **Invoice/receipt** after successful reservation or purchase

3. ⚡ Performance Requirements

- Homepage should load in **under 3 seconds**
- Car availability check should respond within **4 seconds**
- System should support up to **50,000 concurrent users**

4. 🔒 Control (Security Requirements)

- Protect **credit card information** using strong encryption (SSL/TLS)
- Ensure **secure data transmission**
- Protect **customer personal data**
- Implement:
  - Authentication (login system)
  - Authorization (role-based access)
  - Secure payment gateway integration

5. 👥 User Roles

1. Customer

- Browse available cars
- View car details and pricing
- Make reservations or purchases
- Make online payments
- View car details and pricing
- View booking history and invoices

2. Staff (Sales/Support)

(Note: You wrote Barista, but here it should be Sales Staff)

- Manage car listings

2. Staff (Sales/Support)

(Note: You wrote Barista, but here it should be Sales Staff)

- Manage car listings
- Update availability
- Assist customer reservations
- Confirm bookings

3. Admin / Manager

- Manage users (customers & staff)
- Add/edit/delete car information
- Set pricing and promotions
- View reports (sales, bookings)
- Monitor system performance

6. ⚙️ Functional Requirements

Customer Functions

- Register and login
- Search/filter cars
- View car details
- Check availability
- Calculate total price
- Make reservation
- Make payment
- Receive invoice

Staff Functions

- Add/edit car details
- Update car availability
- Manage reservations
- Customer support

Admin Functions

- Full system control
- Manage all users
- Generate reports
- Manage pricing and inventory

7. 🧱 Non-Functional Requirements

- **Performance:** Fast loading and response times
- **Security:** Data encryption, secure login
- **Scalability:** Handle high number of users
- **Usability:** Easy-to-use interface
- **Reliability:** System uptime ≥ 99%
- **Compatibility:** Works on mobile & desktop browsers

8. 🚀 Optional Features (Advanced)

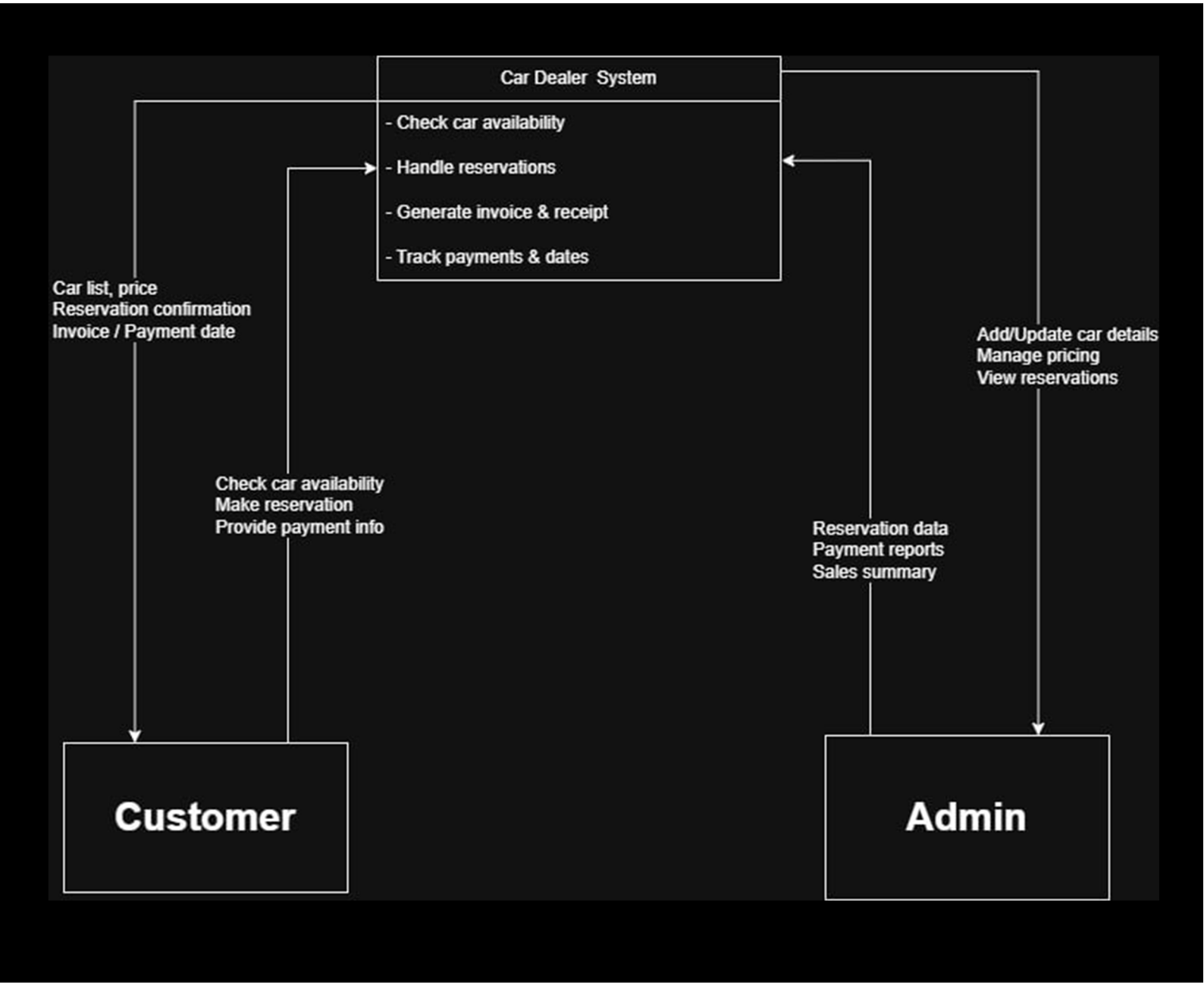
- 🔍 Advanced search (filter by price, brand, fuel type)
- 📊 Analytics dashboard for admin

- 📱 Mobile app version
- ⭐ Customer reviews & ratings
- 📅 Booking calendar
- 🔔 Email/SMS notifications
- 💳 Multiple payment methods (ABA, Visa, MasterCard, etc.)
- 🤖 Chatbot support

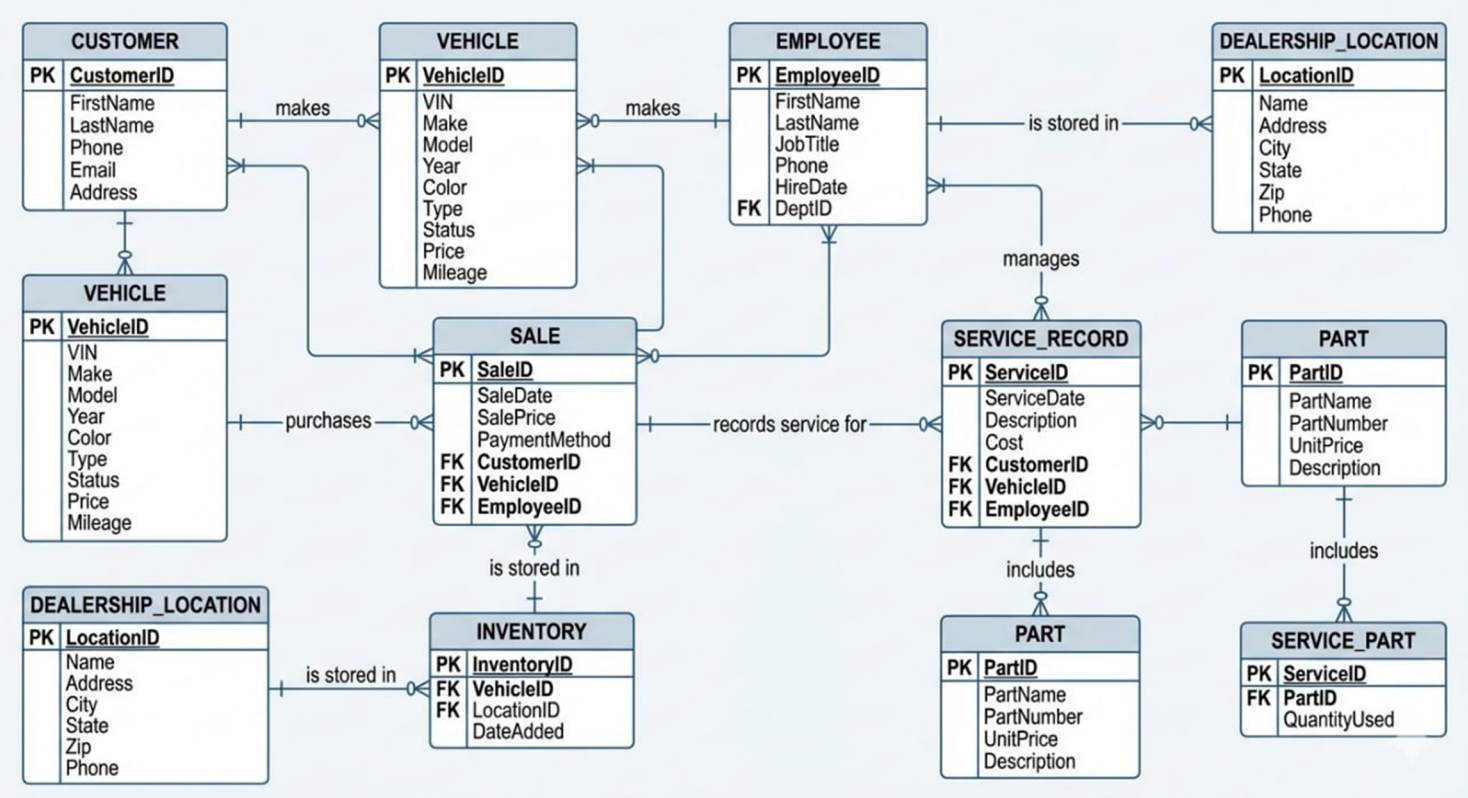
9. 📝 Notes (Fixes from your version)

- “reverse a car” → should be reserve a car
- “Barista” → replaced with **Sales Staff**
- Improved grammar and structure for professional use

Data Flow Diagram (ERD)



Entity Relationship Diagram (ERD)



Data Dictionary

1. Vehicle Type Table

Field Name	Data Type	Description
type_id	INT (PK)	Unique ID for vehicle type
type_name	VARCHAR(100)	Type name (SUV, Sedan, Truck, etc.)
description	VARCHAR(255)	Details about vehicle type

2. Brand Table

Field Name	Data Type	Description
brand_id	INT (PK)	Unique ID for brand
brand_name	VARCHAR(100)	Brand name (Toyota, Ford, etc.)
country	VARCHAR(100)	Country of origin

3. Vehicle Table

Field Name	Data Type	Description
vehicle_id	INT (PK)	Unique vehicle ID
model	VARCHAR(100)	Vehicle model
year	INT	Manufacture year
price	DECIMAL(10,2)	Vehicle price
stock_qty	INT	Available quantity
brand_id	INT (FK)	Reference to Brand Table
type_id	INT (FK)	Reference to Vehicle Type Table

4. Customer Table

Field Name	Data Type	Description
customer_id	INT (PK)	Unique customer ID
name	VARCHAR(100)	Customer full name
phone	VARCHAR(20)	Contact number
email	VARCHAR(100)	Email address
address	VARCHAR(255)	Customer address

5. Reservation Table

Field Name	Data Type	Description
reservation_id	INT (PK)	Unique reservation ID
customer_id	INT (FK)	Reference to Customer
vehicle_id	INT (FK)	Reserved vehicle
reservation_date	DATE	Date of reservation
status	VARCHAR(50)	Pending / Confirmed / Cancelled

6. Payment Table

Field Name	Data Type	Description
payment_id	INT (PK)	Unique payment ID
reservation_id	INT (FK)	Related reservation
payment_date	DATE	Date of payment
amount	DECIMAL(10,2)	Total paid
payment_method	VARCHAR(50)	Cash / Card / Online

7. Invoice Table

Field Name	Data Type	Description
invoice_id	INT (PK)	Unique invoice ID
reservation_id	INT (FK)	Linked reservation
issue_date	DATE	Invoice date
total_amount	DECIMAL(10,2)	Total cost

8. Admin Table

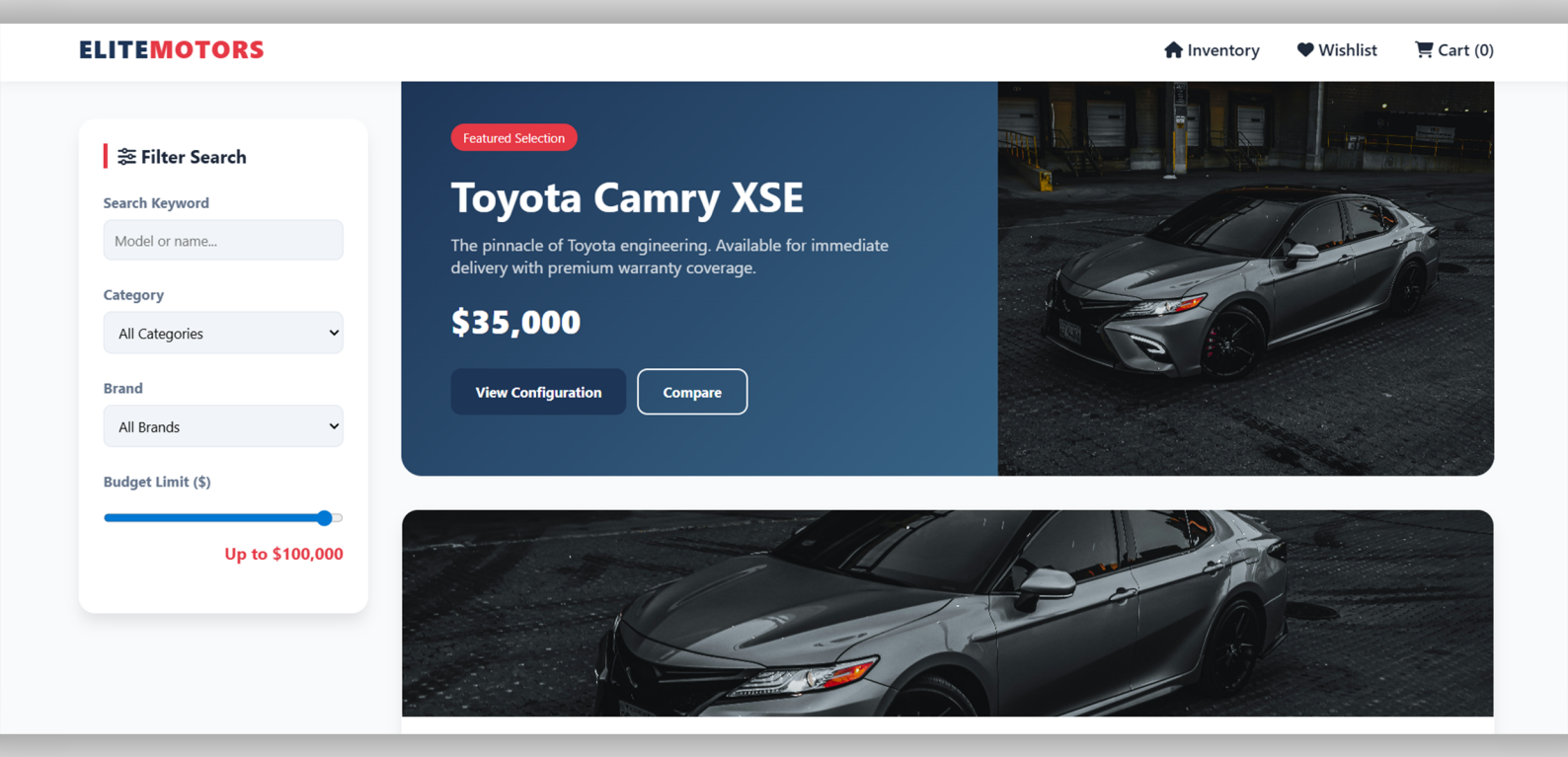
Field Name	Data Type	Description
admin_id	INT (PK)	Unique admin ID
username	VARCHAR(100)	Login username
password	VARCHAR(100)	Login password
role	VARCHAR(50)	Admin role



User Interface

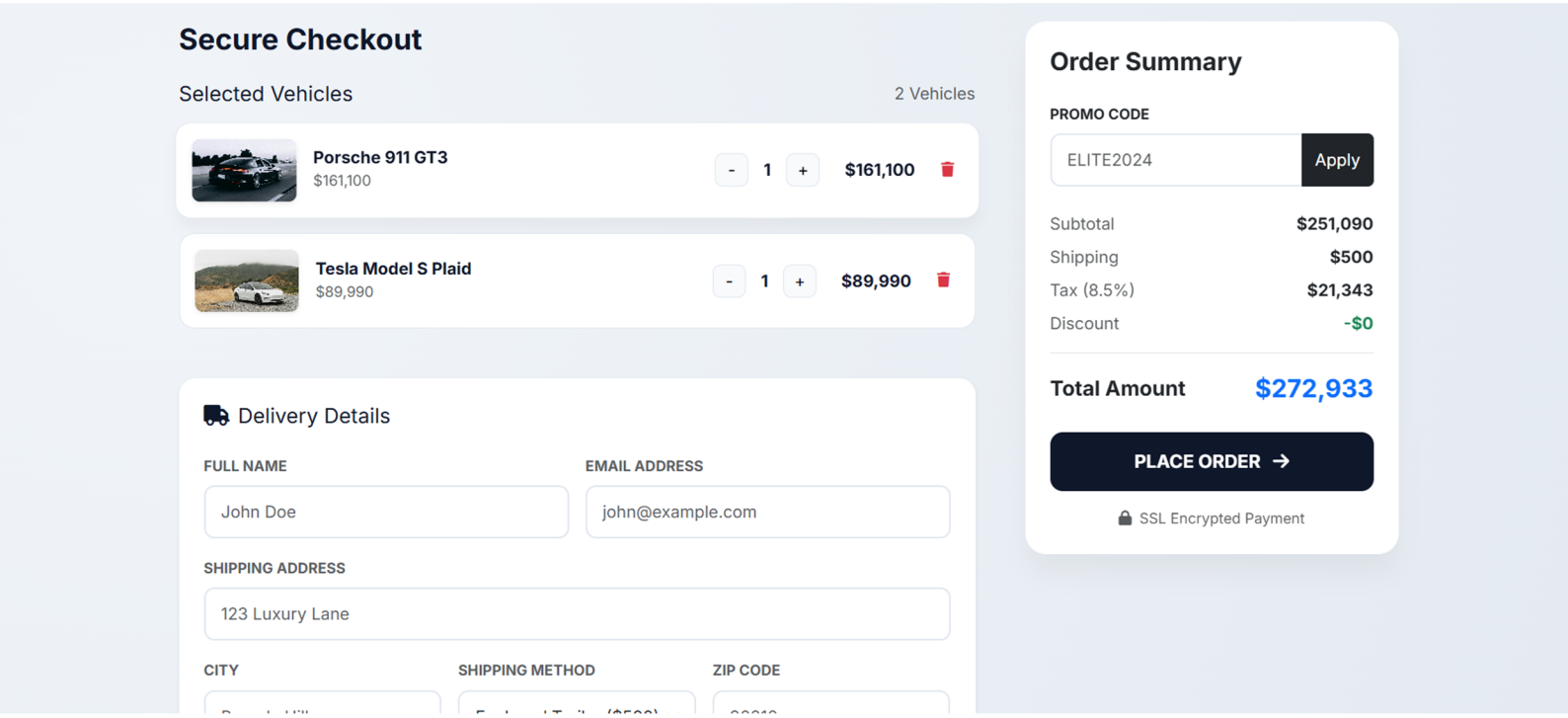
- **Product List**

  A Product List shows or keeps track of products and their details for browsing or management.



- **Check Out**

  Check Out is the process where a customer reviews their selected items, provides payment, and completes the purchase.



Test Case

Case ID	Test Scenario	Test Steps	Test Data	Expected Results	Actual Results	Status
CD-001	Customer views available cars	1. Login as customer 2. Navigate to "Available Cars" page	Customer credentials	List of all available cars is displayed with images, price, and details	TBD	TBD
CD-002	Customer searches car by brand	1. Go to search bar 2. Enter brand "Toyota" 3. Click search	Brand = Toyota	Only Toyota cars are displayed	TBD	TBD
CD-003	Admin adds new car	1. Login as admin 2. Navigate to "Add Car" 3. Enter car details 4. Save	Car: Model=Corolla, Brand=Toyota, Price=$25,000, Stock=5	Car is added successfully and visible in inventory	TBD	TBD
CD-004	Customer makes a reservation	1. Login as customer 2. Select a car 3. Click "Reserve" 4. Confirm reservation	Customer ID=101, Car ID=5	Reservation is successful; reservation confirmation displayed; inventory updated	TBD	TBD
CD-005	Admin generates invoice	1. Login as admin 2. Navigate to "Reservations" 3. Select reservation 4. Click "Generate Invoice"	Reservation ID=201	Invoice is generated with correct customer, car, price, and date	TBD	TBD
CD-006	Customer cancels reservation	1. Login as customer 2. Navigate to "My Reservations" 3. Select reservation 4. Click "Cancel"	Reservation ID=202	Reservation is canceled; car inventory updated; confirmation displayed	TBD	TBD
CD-007	Admin updates car price	1. Login as admin 2. Navigate to "Edit Car" 3. Update price to $27,000 4. Save	Car ID=5, New Price=$27,000	Price is updated; reflected in car listing	TBD	TBD

Product Backlog

ID	Feature / User Story	Estimation (h)	Description	Priority	Sprint	Story Points	Status
Customer Module							
1	As a customer, I would like to view the homepage so that I can see featured cars and promotions.	6h	Display homepage	High	Sprint 1	3	To Do
2	As a customer, I need to view product details so that I can see car info.	6h	Show car details	High	Sprint 1	3	To Do
3	As a customer, I would like to view ratings and comments so that I can read reviews.	4h	Show ratings & comments	Medium	Sprint 2	2	To Do
4	As a customer, I can add products to cart so that I can buy cars.	4h	Add to cart	High	Sprint 1	2	To Do
Account Module							
5	As a customer, I would like to login or create an account so that I can manage my purchases.	8h	Login & register	High	Sprint 1	3	To Do
Cart Module							
6	As a customer, I need to check my cart so that I can review items before checkout.	4h	View cart	High	Sprint 1	2	To Do
Order Module							
7	As a customer, I would like to enter payment details so that I can pay securely.	6h	Payment input	High	Sprint 2	3	To Do
8	As a customer, I can complete checkout so that my order is processed.	10h	Checkout process	High	Sprint 2	5	To Do
9	As a customer, I want the ability to navigate back home so that I can continue browsing.	2h	Navigation feature	Low	Sprint 1	1	To Do
Order Module							
10	As a customer, I would like to view current orders so that I can track progress.	4h	Track orders	Medium	Sprint 2	2	To Do
11	As a customer, I need to view past orders so that I can see order history.	4h	Order history	Medium	Sprint 2	2	To Do

Management Module							
12	As an employee, I would like to view orders so that I can manage them.	6h	View orders	High	Sprint 2	3	To Do
13	As an employee, I need to check revenue so that I can track performance.	6h	Revenue report	Medium	Sprint 3	3	To Do
14	As an employee, I can confirm orders so that processing can start.	4h	Confirm order	High	Sprint 2	2	To Do
15	As an employee, I would like to assign orders to sellers so that they can fulfill them.	4h	Assign orders	High	Sprint 2	2	To Do
Management Module							
16	As an admin, I need to review and confirm orders so that workflow is controlled.	6h	Admin approval	Medium	Sprint 3	3	To Do
Seller Module							
17	As a seller, I would like to receive orders so that I know what to prepare.	4h	Receive order	High	Sprint 2	2	To Do
18	As a seller, I need to prepare items so that they are ready for delivery.	6h	Prepare item	Medium	Sprint 3	3	To Do
Delivery & Feedback Module							
19	As a customer, I would like to receive my product so that I complete the purchase.	4h	Delivery	Low	Sprint 2	2	To Do
20	As a customer, I can submit feedback so that I can share my experience.	4h	Feedback system	Medium	Sprint 2	2	To Do

Sprint = 2 weeks, 4 hours × 14 days = 56 hours, Works = 49 hours, Meeting = 7 hours

Sprint Backlog

ID	Story ID	Description	Response	Start Date	Finish	Estimation (h)	Status
As a customer, I would like to view the homepage so that I can see featured cars and promotions							
1	US-01	Design homepage layout Add sample promotions Verify functionality Finalize UI				6	To Do
As a customer, I need to view product details so that I can see car information							
2	US-02	Build product detail UI Populate mock product data Run tests Finalize layout				6	To Do
As a customer, I would like to view ratings and comments so that I can read reviews							
3	US-03	Develop ratings & comments section Input test data Verify functionality Complete layout				4	To Do
As a customer, I can add products to the cart so that I can purchase cars							
4	US-04	Build “Add to Cart” functionality Insert mock items Test workflow Finalize interface				4	To Do
As a customer, I would like to log in or create an account so that I can manage my purchases							
5	US-05	Develop login & registration forms Add sample accounts Test login flow Finalize layout				8	To Do

As a customer, I need to check my cart so that I can review items before checkout				
6	US-06	Build cart overview UI Insert mock items Verify cart updates Finalize design	4	To Do
As a customer, I would like to enter payment details so that I can complete my payment securely				
7	US-07	Create payment form Add test data Check validation Finalize interface	6	To Do
As a customer, I can complete checkout so that my order is processed successfully				
8	US-08	Build checkout process Add sample orders Run workflow tests Finalize layout	10	To Do
As a customer, I want the ability to navigate back to the homepage so that I can continue browsing				
9	US-09	Add navigation feature Insert mock links Test back navigation Finalize UI	2	To Do
As a customer, I would like to view current orders so that I can track their progress				
10	US-10	Build current orders page Add sample orders Test tracking Finalize layout	4	To Do
As a customer, I need to view past orders so that I can review my order history				
11	US-11	Develop past orders interface Insert mock history Verify workflow Finalize design	4	To Do
As an employee, I would like to view orders so that I can manage them effectively				
12	US-12	Build employee order management page Add sample orders Test workflow Finalize UI	6	To Do
As an employee, I need to check revenue so that I can monitor business performance				
13	US-13	Create revenue report interface Input test data Verify calculations Finalize layout	6	To Do
As an employee, I can confirm orders so that processing can begin				
14	US-14	Build order confirmation module Add mock orders Run tests Finalize design	4	To Do
As an employee, I would like to assign orders to sellers so that they can fulfill them				
15	US-15	Create order assignment feature Add test data Verify workflow Finalize interface	4	To Do
As an admin, I need to review and confirm orders so that the workflow is properly controlled				
16	US-16	Build admin approval page Insert mock orders Test workflow Finalize layout	6	To Do

As a seller, I would like to receive orders so that I know what items to prepare				
17	US-17	Create seller order inbox Add test orders Verify receipt Finalize UI	4	To Do
As a seller, I need to prepare items so that they are ready for delivery				
18	US-18	Build item preparation interface Add sample items Run workflow tests Finalize layout	6	To Do
As a customer, I would like to receive my product so that I can complete the purchase process				
19	US-19	Create delivery tracking page Add sample deliveries Verify updates Finalize UI	4	To Do
As a customer, I can submit feedback so that I can share my experience				
20	US-20	Build feedback form Input mock reviews Test submission Finalize design	4	To Do

Summary

Purpose of the Car Dealer System

The Car Dealer system is designed to manage the buying, selling, and management of vehicles in a dealership environment. It serves both customers and dealership staff/admins by providing a digital interface for browsing, purchasing, and managing vehicles.

Key Functions

1. Customer Functions:

- Browse cars on the homepage (featured cars, promotions).
- View detailed product/car information (price, specs, images).
- ETC

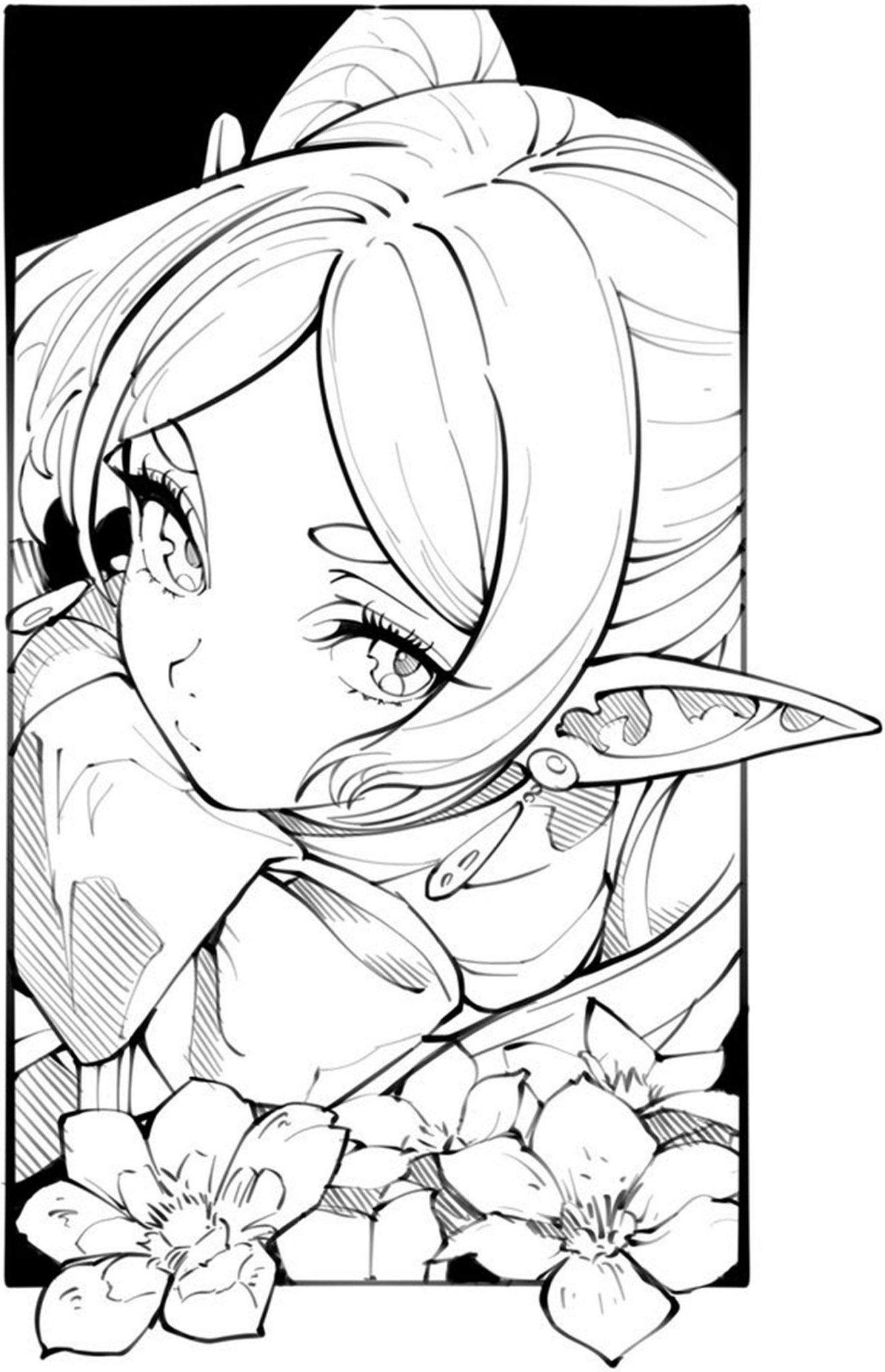
2. Admin / Employee Functions:

- Manage inventory (add/edit/delete vehicles, brands, categories).
- View and process customer orders.
- ETC

3. System Features:

- Product categorization (e.g., by brand, type).
- Image upload and display for cars.
- ETC

In short, the **Car Dealer system** is a **full-stack dealership management** and **e-commerce platform** that allows customers to browse and buy cars online while giving staff tools to smanage inventory and sales efficiently.



CAR DEALER

A4

BOOK

THE BEST

MADE BY CHOU YAMATA